

ICMR-NIN Dietary Guidelines for Indians (2024)

Reference Guide for NutriSage AI - Supporting UN SDG Goal 3: Good Health & Well-Being

1. Executive Overview

The 2024 edition of the Dietary Guidelines for Indians (DGI) has been formulated by the Indian Council of Medical Research (ICMR) - National Institute of Nutrition (NIN) in Hyderabad. The guidelines address India's evolving nutritional profile and the dual burden of malnutrition. With a surge in non-communicable diseases (NCDs) such as type-2 diabetes, hypertension, and cardiovascular diseases, the guidelines serve as a clinical and educational framework to foster balanced physiological development and long-term health.

2. The 17 Core Dietary Guidelines

- **Guideline 1: Consume a Variety of Foods:** Ensure a balanced diet by including diverse food groups such as cereals, millets, pulses, milk, fresh vegetables, fruits, and healthy fats.
- **Guideline 2: Maternal Nutrition:** Ensure extra food, micronutrient intake, and comprehensive healthcare during pregnancy and lactation to support fetal and infant growth.
- **Guideline 3: Infant Feeding:** Promote exclusive breastfeeding for the first six months. Introduce nutrient-dense complementary foods after six months while continuing breastfeeding.
- **Guideline 4: Childhood & Adolescence:** Feed optimal, balanced diets to infants, young children, and adolescents to support rapid growth and cognitive development.
- **Guideline 5: Eat Vegetables & Fruits:** Eat plenty of fresh vegetables, especially green leafy vegetables, and seasonal fruits to ensure adequate micronutrients and antioxidants.
- **Guideline 6: Natural Protein Sources:** Obtain high-quality protein from combinations of natural foods (pulses, legumes, dairy, nuts, lean meats) rather than relying on commercial protein supplements.
- **Guideline 7: Limit processed & HFSS foods:** Minimize the intake of high fat, salt, and sugar (HFSS) foods, ultra-processed foods (UPFs), and carbonated beverages.
- **Guideline 8: Moderation of Fats & Oils:** Limit consumption of cooking oils and fats. Choose a mix of vegetable oils and strictly avoid repeated heating/re-using of oils to prevent toxic trans-fat formation.
- **Guideline 9: Prevent Obesity:** Maintain a healthy body weight. Combine regular physical activity (minimum 30-45 minutes/day) with calorie-conscious dietary patterns.
- **Guideline 10: Hydration:** Drink safe, clean water in adequate amounts daily (typically 2 to 2.5 liters) to maintain physiological homeostasis.
- **Guideline 11: Restrict Salt & Sugar:** Restrict salt intake to less than 5g per day and minimize added sugars to prevent hypertension, renal stress, and type-2 diabetes.
- **Guideline 12: Nutrient Retention:** Adopt healthy food pre-cooking (soaking, sprouting) and cooking practices (steaming, boiling, minimal frying) to preserve essential nutrients.
- **Guideline 13: Read Food Labels:** Check nutrition facts panels and ingredients lists on packaged foods to make informed, healthier purchasing choices.

- **Guideline 14: Food Safety & Hygiene:** Ensure food hygiene, sanitation of food preparation areas, and safe storage practices to prevent foodborne pathogens.
- **Guideline 15: Elderly Care:** Support the elderly with nutrient-dense, easily chewable, and easily digestible foods rich in calcium, iron, and fiber.
- **Guideline 16: Lifestyle & Stress:** Avoid harmful habits such as tobacco and alcohol consumption. Practice mindfulness and active stress management.
- **Guideline 17: Sustainable Diets:** Choose locally sourced, seasonal, and environmentally sustainable foods that support regional biodiversity and agricultural practices.

3. Specific Indian Food Recommendations & Calorie Splits

Cereal-to-Calorie Limit:

The guidelines recommend limiting total cereal intake to **no more than 45%** of total daily energy intake. This is intended to prevent excessive carbohydrate loading, which is a major contributor to insulin resistance and type-2 diabetes in the Indian population.

Plant-Based Protein Options:

To meet protein requirements (especially for vegetarians), combine cereals with pulses (ratio of 3:1 or 4:1) or consume high-protein vegetarian options such as:

- **Moong Dal & Chana Dal:** Traditional split beans containing 22-24g protein per 100g.
- **Paneer & Tofu:** Excellent sources of complete proteins and calcium.
- **Chickpeas & Lentils:** High in dietary fiber and essential amino acids.
- **Sprouts:** Increased bioavailability of vitamins and minerals through germination.

Millets & Low Glycemic Index (GI) Foods:

Replace refined grains (maida, white rice) with whole grains and millets:

- **Ragi (Finger Millet):** Exceptionally rich in calcium and dietary fiber, crucial for bone health and glycemic control.
- **Bajra (Pearl Millet) & Jowar (Sorghum):** Rich in iron, magnesium, and resistant starch, which improves insulin sensitivity and prevents blood glucose spikes.
- **Oats & Barley:** Contains beta-glucans which actively lower LDL cholesterol levels.

Cardiovascular Health & Dietary Fiber:

To support SDG 3.4 (reducing NCD mortality), adults must consume a minimum of **30 grams of dietary fiber per day**. High-fiber diets reduce cholesterol absorption, lower blood pressure, and regulate cardiovascular strain.

NutriSage AI Calorie & Macro Target Split Reference

Goal	Carbohydrates (%)	Protein (%)	Fat (%)	Focus / Clinical Rationale
Weight Loss	40%	35%	25%	Preserve lean mass, reduce insulin response
Maintenance	50%	20%	30%	Energy homeostasis, balanced metabolic profile
Weight Gain	55%	25%	20%	Support muscular hypertrophy, glycogen replenishment